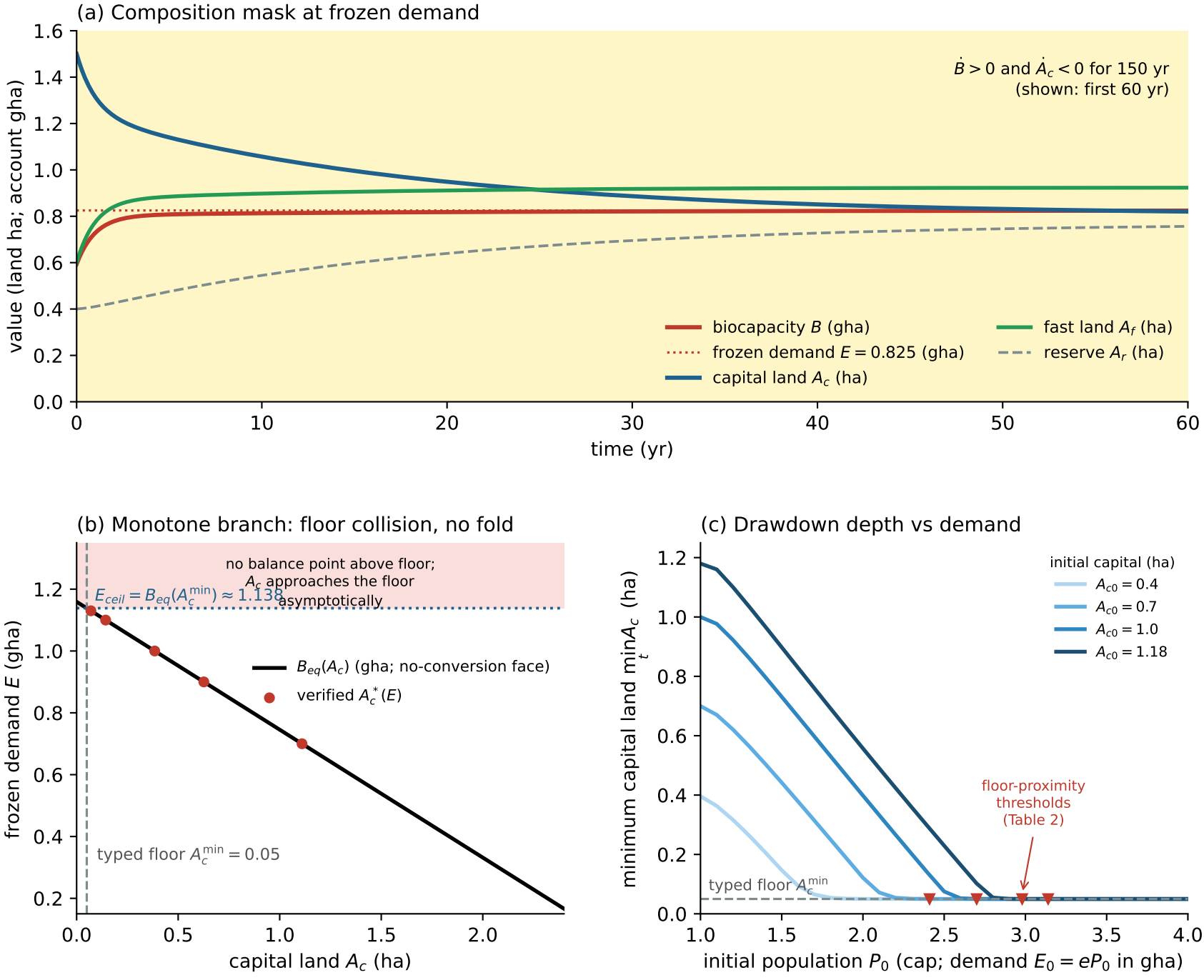
(a) Composition mask at frozen demand
$\dot{B} > 0$ and $\dot{A}_c < 0$ for 150 yr
(shown: first 60 yr)
biocapacity B (gha)
frozen demand $E = 0.825$ (gha)
capital land A_c (ha)
fast land A_f (ha)
reserve A_r (ha)
value (land ha; account gha)
time (yr)
(b) Monotone branch: floor collision, no fold
no balance point above floor;
A_c approaches the floor
asymptotically
$E_{ceil} = B_{eq}(A_c^{min}) \approx 1.138$
$B_{eq}(A_c)$ (gha; no-conversion face)
verified $A_c^*(E)$
typed floor $A_c^{min} = 0.05$
frozen demand E (gha)
capital land A_c (ha)
(c) Drawdown depth vs demand
initial capital (ha)
$A_{c0} = 0.4$
$A_{c0} = 0.7$
$A_{c0} = 1.0$
$A_{c0} = 1.18$
floor-proximity
thresholds
(Table 2)
typed floor A_c^{min}
minimum capital land $\min_t A_c$ (ha)
initial population P_0 (cap; demand $E_0 = eP_0$ in gha)